Chapter 6
Multicollinearity

A basic assumption in the multiple linear regression model is that the rank of the matrix of observations on the explanatory variables equals the number of explanatory variables. In other words, such a matrix has full column rank. This, in turn, implies that all the explanatory variables are independent, i.e., there is no linear relationship among them. It is said that the explanatory variables are orthogonal.

In many practical situations, the explanatory variables may not remain independent for various reasons. The situation where the explanatory variables are highly intercorrelated is referred to as **multicollinearity.**

Consider the multiple regression model

$$y = \underset{n\times k}{X}\ \underset{k\times 1}{\beta} + \underset{n\times 1}{\varepsilon},\ \varepsilon \sim N(0, \sigma^2 I)$$

with k explanatory variables $X_1, X_2, ..., X_k$ with usual assumptions, including $\text{Rank}(X) = k$.

Assume the observations on all $X_i's$ and $y_i's$ are centered and scaled to unit length. So

- $X'X$ becomes a $k \times k$ matrix of correlation coefficients between the explanatory variables and
- $X'y$ becomes a $k \times 1$ vector of correlation coefficients between explanatory and study variables.

Let $X = [X_1, X_2, ..., X_k]$ where X_j is the jth column of X denoting the *n* observations on X_j. The column vectors $X_1, X_2, ..., X_k$ are linearly dependent if there exists a set of constants $\ell_1, \ell_2, ..., \ell_k$, not all zero, such that

$$\sum_{j=1}^{k} \ell_j X_j = 0.$$

If this holds exactly for a subset of the $X_1, X_2, ..., X_k$, then $\text{rank}(X'X) < k$. Consequently $(X'X)^{-1}$ does not exist. If the condition $\sum_{j=1}^{k} \ell_j X_j = 0$ is approximately true for some subset of $X_1, X_2, ..., X_k$, then there will be a near-linear dependency in $X'X$. In such a case, the multicollinearity problem exists. It is also said that $X'X$ becomes **ill-conditioned.**

Source of multicollinearity

1. Method of data collection

It is expected that the data will be collected across the entire cross-section of variables. It may happen that the data are collected over a subspace of the explanatory variables in which the variables are linearly dependent. For example, sampling is done only over a limited range of explanatory variables in the population.

2. Model and population constraints

There may exist some constraints on the model or on the population from which the sample is drawn. The sample may be generated from that part of the population that has linear combinations.

3. Existence of identities or definitional relationships

There may be relationships among the variables, arising from their definitions or from identity relations among them. For example, if data are collected on variables such as income, savings, and expenditures, then income = savings + expenditures. Such a relationship will not change even when the sample size increases.

4. Imprecise formulation of the model

The formulation of the model may unnecessarily be complicated. For example, the quadratic (or polynomial) terms or cross-product terms may appear as explanatory variables. For example, let there be 3 variables X_1, X_2 and X_3, so $k = 3$. Suppose their cross-product terms X_1X_2, X_2X_3 and X_1X_3 are also added. Then k rises to 6.

5. An over-determined model

Sometimes, due to over-enthusiasm, a large number of variables are included in the model to make it more realistic. Consequently, the number of observations (n) becomes smaller than the number of explanatory variables (k). Such a situation can arise in medical research, where the number of patients may be small, but the data are collected across many variables. In another example, if there is time-series data spanning 50 years on consumption patterns, it is expected that consumption patterns will change over that period. So a better option is to choose a smaller number of variables, and hence it results in $n < k$.

Consequences of multicollinearity

To illustrate the consequences of the presence of multicollinearity, consider a model

$$y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon,\ E(\varepsilon) = 0,\ V(\varepsilon) = \sigma^2 I$$

where x_1, x_2 and y are scaled to length unity.

The normal equation $(X'X)b = X'y$ in this model becomes

$$\begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} r_{1y} \\ r_{2y} \end{pmatrix}$$

where r is the correlation coefficient between x_1 and x_2; r_{jy} is the correlation coefficient between x_j and $y(j = 1,2)$ and $b = (b_1, b_2)'$ is the OLSE of β.

$$(X'X)^{-1} = \left(\frac{1}{1-r^2}\right)\begin{bmatrix} 1 & -r \\ -r & 1 \end{bmatrix}$$

$$\Rightarrow b_1 = \frac{r_{1y} - r\, r_{2y}}{1-r^2}$$

$$b_2 = \frac{r_{2y} - r\, r_{1y}}{1-r^2}.$$

So the covariance matrix is $V(b) = \sigma^2 (X'X)^{-1}$

$$\Rightarrow Var(b_1) = Var(b_2) = \frac{\sigma^2}{1-r^2}$$

$$Cov(b_1, b_2) = -\frac{r\sigma^2}{1-r^2}.$$

If x_1 and x_2 are uncorrelated, then $r = 0$ and

$$X'X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\text{rank}(X'X) = 2.$$

If x_1 and x_2 are perfectly correlated, then $r = \pm 1$ and $\text{rank}(X'X) = 1.$

If $r \to \pm 1,$ then $Var(b_1) = Var(b_2) \to \infty$.

So if variables are perfectly collinear, the variance of OLSEs becomes large. This indicates highly unreliable estimates, and this is an inadmissible situation.

Consider the following result

r	0.99	0.9	0.1	0
$Var(b_1) = Var(b_2)$	$50\sigma^2$	$5\sigma^2$	$1.01\sigma^2$	σ^2

The standard errors of b_1 and b_2 rise sharply as $r \to \pm 1$ and they break down at $r = \pm 1$ because $X'X$ becomes non-singular.

- If r is close to 0, then multicollinearity does not harm, and it is termed as **non-harmful multicollinearity**.
- If r is close to +1 or -1 then multicollinearity inflates the variance, and it rises terribly. This is termed as **harmful multicollinearity.**

There is no clear-cut boundary to distinguish between harmful and non-harmful multicollinearity. Generally, if r is low, the multicollinearity is considered as non-harmful, and if r is high, multicollinearity is considered harmful.

In cases of near or high multicollinearity, the following consequences may occur.

1. The OLSE remains an unbiased estimator of β**,** but its sampling variance becomes very large. So, OLSE becomes imprecise, and the BLUE property no longer holds.
2. Due to large standard errors, the regression coefficients may not appear significant. Consequently, essential variables may be dropped.

   For example, to test $H_0 : \beta_1 = 0,$ we use $t-$ratio as

   $$t_0 = \frac{b_1}{\sqrt{Var(b_1)}}.$$

   Since $Var(b_1)$ is large, so t_0 is small and consequently H_0 is more often accepted.

   Thus, harmful multicollinearity can lead to the deletion of important variables.

3. Due to large standard errors, a large confidence region may arise. For example, the confidence interval is given by $\left(b_1 \pm t_{\frac{\alpha}{2}, n-1} \sqrt{Var(b_1)} \right)$. When $Var(b_1)$ it becomes large, then the confidence interval becomes wider.
4. The OLSE may be sensitive to small changes in the values of explanatory variables. If some observations are added or dropped, the OLSE may change considerably in magnitude and sign. Ideally, OLSE should not change when variables are added or removed. Thus, OLSE loses stability and robustness.

When the number of explanatory variables is more than two, say k as $X_1, X_2, ..., X_k$ then the jth diagonal element of $C = (X'X)^{-1}$ is

$$C_{jj} = \frac{1}{1 - R_j^2}$$

where R_j^2 is the multiple correlation coefficient or the coefficient of determination from the regression of X_j on the remaining $(k-1)$ explanatory variables.

If X_j is highly correlated with any subset of the other $(k-1)$ explanatory variables then R_j^2 is high and close to 1. Consequently, the variance of jth OLSE $Var(b_j) = C_{jj}\sigma^2 = \frac{\sigma^2}{1 - R_j^2}$ becomes very high. The covariance between b_i and b_j will also be large if X_i and X_j are involved in a linear relationship that leads to multicollinearity.

The least-squares estimates b_j become too large in absolute value in the presence of multicollinearity. For example, consider the squared distance between b and β as

$$\begin{aligned} L^2 &= (b-\beta)'(b-\beta) \\ E(L^2) &= \sum_{j=1}^{k} E(b_j - \beta_j)^2 \\ &= \sum_{j=1}^{k} Var(b_j) \\ &= \sigma^2 tr(X'X)^{-1}. \end{aligned}$$

The trace of a matrix is the same as the sum of its eigenvalues. If $\lambda_1, \lambda_2, ..., \lambda_k$ are the eigenvalues of $(X'X)$, then $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, ..., \frac{1}{\lambda_k}$ are the eigenvalues of $(X'X)^{-1}$ and hence

$$E(L^2) = \sigma^2 \sum_{j=1}^{k} \frac{1}{\lambda_j}, \ \lambda_j > 0.$$

If $(X'X)$ is ill-conditioned due to multicollinearity, then at least one eigenvalue will be small. So the distance between b and β may also be substantial. Thus

$$E(L^2) = E(b-\beta)'(b-\beta)$$
$$\sigma^2 tr(X'X)^{-1} = E(b'b - 2b'\beta + \beta'\beta)$$
$$\Rightarrow E(b'b) = \sigma^2 tr(X'X)^{-1} + \beta'\beta$$
$$\Rightarrow b \text{ is generally longer than } \beta$$
$$\Rightarrow \text{OLSE is too large in absolute value.}$$

The least-squares method produces incorrect parameter estimates when multicollinearity is present. This does not imply that the fitted model provides wrong predictions, either. If the predictions are confined to *x*-space and there is no harmful multicollinearity, then the predictions are satisfactory.

Multicollinearity diagnostics

An important question arises about how to diagnose multicollinearity in the data using the given sample information. Several diagnostic measures are available, each based on a particular approach. It is difficult to say which diagnostic is best or ultimate. Some of the popular and important diagnostics are described in more detail. The detection of multicollinearity involves 3 aspects:

(i) Determining its presence.
(ii) Determining its severity.
(iii) Determining its form or location.

1. Determinant of $X'X$ $\left(|X'X|\right)$

This measure is based on the fact that the matrix $X'X$ becomes ill-conditioned in the presence of multicollinearity. The value of the determinant of $X'X$, i.e., $|X'X|$ declines as the degree of multicollinearity increases.

If $\text{Rank}(X'X) < k$ then $|X'X|$ will be singular and so $|X'X| = 0$. So, as $|X'X| \to 0$, the degree of multicollinearity increases, and it becomes exact or perfect at $|X'X| = 0$. Thus $|X'X|$ serves as a measure of multicollinearity and $|X'X|$=0 indicates the presence of perfect multicollinearity.

Limitations

This measure has the following limitations

(i) It is not bounded as $0 < |X'X| < \infty$.

(ii) It is affected by the dispersion of explanatory variables. For example, if $k = 2,$ then

$$|X'X| = \begin{vmatrix} \sum_{i=1}^{n} x_{1i}^2 & \sum_{i=1}^{n} x_{1i}x_{2i} \\ \sum_{i=1}^{n} x_{2i}x_{1i} & \sum_{i=1}^{n} x_{2i}^2 \end{vmatrix}$$
$$= \left(\sum_{i=1}^{n} x_{1i}^2\right)\left(\sum_{i=1}^{n} x_{2i}^2\right)\left(1 - r_{12}^2\right)$$

where r_{12} is the correlation coefficient between x_1 and x_2. So $|X'X|$ depends on the explanatory variable's correlation coefficient and variability. If explanatory variables have very low variability, then $|X'X|$ may tend towards zero, indicating the presence of multicollinearity, but this is not the case.

(iii) It gives no idea about the relative effects on individual coefficients. If multicollinearity is present, then it will not indicate which variable in $|X'X|$ is causing multicollinearity and is hard to determine.

2. Inspection of correlation matrix

The inspection of off-diagonal elements r_{ij} in $X'X$ gives an idea about the presence of multicollinearity. If X_i and X_j are nearly linearly dependent, then $|r_{ij}|$ will be close to 1. Note that the observations in X are standardized in the sense that each observation is subtracted from the mean of that variable and divided by the square root of the corrected sum of squares of that variable.

When more than two explanatory variables are considered, and if they are involved in near-linear dependency, then it is not necessary that any of the r_{ij} will be large. Generally, a pairwise inspection of correlation coefficients is insufficient to detect multicollinearity in the data.

3. Determinant of correlation matrix

Let D be the determinant of the correlation matrix, then $0 \le D \le 1$.

If $D = 0$, then it indicates the existence of exact linear dependence among explanatory variables.

If $D = 1$ then the columns of X matrix are orthonormal.

Thus, a value close to 0 indicates a high degree of multicollinearity. Any value of D between 0 and 1 indicates the degree of multicollinearity.

Limitation

It provides no information on the number of linear dependencies among the explanatory variables.

Advantages over $|X'X|$

(i) It is a bounded measure, $0 \le D \le 1$.

(ii) It is not affected by the dispersion of explanatory variables. For example, when $k = 2$,

$$|X'X| = \begin{vmatrix} \sum_{i=1}^{n} x_{1i}^2 & \sum_{i=1}^{n} x_{1i}x_{2i} \\ \sum_{i=1}^{n} x_{1i}x_{2i} & \sum_{i=1}^{n} x_{2i}^2 \end{vmatrix} = (1 - r_{12}^2).$$

4. Measure based on partial regression

A measure of multicollinearity can be obtained from the coefficients of determination based on partial regression. Let R^2 be the coefficient of determination in the full model, i.e., based on all explanatory variables and R_i^2 be the coefficient of determination in the model when the ith explanatory variable is dropped, $i = 1, 2, ..., k$, and $R_L^2 = Max(R_1^2, R_2^2, ..., R_k^2)$.

Procedure

(i) Drop one of the explanatory variables among k variables, say X_1.

(ii) Run a regression of y over the rest of the $(k-1)$ variables $X_2, X_3, ..., X_k$.

(iii) Calculate R_1^2.

(iv) Similarly, calculate $R_2^2, R_3^2, ..., R_k^2$.

(v) Find $R_L^2 = Max(R_1^2, R_2^2, ..., R_k^2)$.

(vi) Determine $R^2 - R_L^2$.

The quantity $(R^2 - R_L^2)$ provides a measure of multicollinearity. If multicollinearity is present, R_L^2 will be high. The higher the degree of multicollinearity, the higher the value of R_L^2. So in the presence of multicollinearity, $(R^2 - R_L^2)$ be low.

Thus if $(R^2 - R_L^2)$ is close to 0, it indicates a high degree of multicollinearity.

Limitations

(i) It gives no information about the underlying relations of explanatory variables, i.e., how many relationships are present or how many explanatory variables are responsible for the multicollinearity.

(ii) A small value of $(R^2 - R_L^2)$ may occur due to poor model specification, and in such situations, multicollinearity may be inferred.

5. Variance inflation factors (*VIF*)

The matrix $X'X$ becomes ill-conditioned due to multicollinearity in the data. So the diagonal elements of $C = (X'X)^{-1}$ helps in the detection of multicollinearity. If R_j^2 denotes the coefficient of determination obtained when X_j is regressed on the remaining $(k-1)$ variables excluding X_j, then the jth diagonal element of C is

$$C_{jj} = \frac{1}{1-R_j^2}.$$

If X_j is nearly orthogonal to the remaining explanatory variables, then R_j^2 is small and consequently C_{jj} is close to 1.

If X_j is nearly linearly dependent on a subset of remaining explanatory variables, then R_j^2 is close to 1 and consequently C_{jj} is large.

Since the variance of jth OLSE of β_j is

$$Var(b_j) = \sigma^2 C_{jj}$$

So C_{jj} is the factor by which the variance of b_j increases when the explanatory variables are near-linearly dependent. Based on this concept, the variance inflation factor for the jth explanatory variable is defined as

$$VIF_j = \frac{1}{1-R_j^{2`}}.$$

This is the factor which is responsible for inflating the sampling variance. The combined effect of dependencies among the explanatory variables on the variance of a term is measured by the *VIF* of that term in the model.

One or more large *VIF*s indicate the presence of multicollinearity in the data.

In practice, usually, a $VIF > 5$ or 10 indicates that the associated regression coefficients are poorly estimated because of multicollinearity. If regression coefficients are estimated by OLSE, and its variance is $\sigma^2(X'X)^{-1}$. So, *VIF* indicates that a part of this variance is accounted for by VIF_j.

Limitations

(i) It sheds no light on the number of dependencies among the explanatory variables.

(ii) The rule of $VIF > 5$ or 10 is a rule of thumb which may differ from one situation to another situation.

Another interpretation of VIF_j

The *VIF*s can also be viewed as follows.

The confidence interval of jth OLSE of β_j is given by

$$\left(b \pm \sqrt{\hat{\sigma}^2 C_{jj}}\, t_{\frac{\alpha}{2},n-k} \right).$$

The length of the confidence interval is

$$L_j = 2\sqrt{\hat{\sigma}^2 C_{jj}}\, t_{\frac{\alpha}{2},n-k}.$$

Now consider a situation where X is an orthogonal matrix, i.e., $X'X = I$ so that $C_{jj} = 1,$ sample size is the same as earlier, and the same root mean squares $\left(\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2 \right)$, then the length of the confidence interval becomes

$$L^* = 2\hat{\sigma} t_{\frac{\alpha}{2},n-k-1}.$$

Consider the ratio

$$\frac{L_j}{L^*} = \sqrt{C_{jj}}.$$

Thus $\sqrt{VIF_j}$ indicates the increase in the length of the confidence interval of jth regression coefficient due to the presence of multicollinearity.

6. Condition number and condition index

Let $\lambda_1, \lambda_2, ..., \lambda_k$ be the eigenvalues (or characteristic roots) of $X'X$. Let

$$\lambda_{\max} = Max(\lambda_1, \lambda_2, ..., \lambda_k)$$
$$\lambda_{\min} = Min(\lambda_1, \lambda_2, ..., \lambda_k).$$

The condition number (CN) is defined as

$$CN = \frac{\lambda_{\max}}{\lambda_{\min}},\ 0 < CN < \infty .$$

The small values of characteristic roots indicate the presence of near-linear dependency in the data. The CN provides a measure of spread in the spectrum of characteristic roots of $X'X$.

The condition number provides a measure of multicollinearity.

- If $CN < 100,$ then it is considered as **non-harmful multicollinearity**.
- If $100 < CN < 1000,$ then it indicates that the multicollinearity is moderate to severe (or strong). This range is referred to as **danger level.**
- If $CN > 1000,$ then it indicates a **severe** (or **strong**) **multicollinearity**.

The condition number is based only on two eigenvalues: $\lambda_{\min}$ and $\lambda_{\max}$. Another measure is the condition index, which uses information about other eigenvalues.

The **condition indices** of $X'X$ are defined as

$$C_j = \frac{\lambda_{\max}}{\lambda_j},\ j = 1, 2, ..., k.$$

In fact, the largest $C_j = CN.$

The number of condition indices that are large, say more than 1000, indicate the number of near-linear dependencies in $X'X$.

A limitation of CN and C_j is that they are unbounded measures as $0 < CN < \infty$, $0 < C_j < \infty$.

7. Measure based on characteristic roots and proportion of variances

Let $\lambda_1, \lambda_2, .., \lambda_k$ be the eigenvalues of $X'X$, $\Lambda = diag(\lambda_1, \lambda_2, ..., \lambda_k)$ is a $k \times k$ matrix and V is a $k \times k$ matrix constructed by the eigenvectors of $X'X$. Obviously, V is an orthogonal matrix. Then $X'X$ can be decomposed as $X'X = V\Lambda V'$. Let $V_1, V_2, ..., V_k$ be the column of V. If there is a near-linear dependency in the data, then λ_j is close to zero, and the nature of linear dependency is described by the elements of the associated eigenvector V_j.

The covariance matrix of OLSE is

$$\begin{aligned} V(b) &= \sigma^2 (X'X)^{-1} \\ &= \sigma^2 (V\Lambda V')^{-1} \\ &= \sigma^2 V\Lambda^{-1} V' \\ \Rightarrow Var(b_i) &= \sigma^2 \left(\frac{v_{i1}^2}{\lambda_1} + \frac{v_{i2}^2}{\lambda_2} + ... + \frac{v_{ik}^2}{\lambda_k} \right) \end{aligned}$$

where $v_{i1}, v_{i2}, ..., v_{ik}$ are the elements in V.

The condition indices are

$$C_j = \frac{\lambda_{\max}}{\lambda_j}, \; j = 1, 2, ..., k.$$

Procedure

(i) Find the condition index $C_1, C_2, ..., C_k$.

(ii) (a) Identify those $\lambda_i's$ for which C_j is greater than the danger level 1000.

(b) This gives the number of linear dependencies.

(c) Don't consider those $C_j's$ which are below the danger level.

(iii) For such $\lambda's$ with condition index above the danger level, choose one such eigenvalue, say λ_j.

(iv) Find the value of the proportion of variance corresponding to λ_j in $Var(b_1), Var(b_2), ..., Var(b_k)$ as

$$p_{ij} = \frac{(v_{ij}^2 / \lambda_j)}{VIF_j} = \frac{v_{ij}^2 / \lambda_j}{\sum_{j=1}^{k} (v_{ij}^2 / \lambda_j)}.$$

Note that $\left(\frac{v_{ij}^2}{\lambda_j}\right)$ can be found from the expression

$$Var(b_i) = \sigma^2\left(\frac{v_{i1}^2}{\lambda_1} + \frac{v_{i2}^2}{\lambda_2} + ... + \frac{v_{ik}^2}{\lambda_k}\right)$$

i.e., corresponding to j^{th} factor.

The proportion of variance p_{ij} provides a measure of multicollinearity.

If $p_{ij} > 0.5,$ it indicates that b_i is adversely affected by the multicollinearity, i.e., an estimate of β_i is influenced by the presence of multicollinearity.

It is a good diagnostic tool in the sense that it indicates the presence of harmful multicollinearity and the number of linear dependencies responsible for it. This diagnostic is better than other diagnostics.

The condition indices are also defined by the singular value decomposition of X the matrix as follows:

$$X = UDV'$$

where U is $n \times k$ matrix, V is $k \times k$ matrix, $U'U = I,\ V'V = I,\ D$ is $k \times k$ matrix, $D = diag(\mu_1, \mu_2, ..., \mu_k)$ and $\mu_1, \mu_2, ..., \mu_k$ are the singular values of X, V is a matrix whose columns are eigenvectors corresponding to eigenvalues of $X'X$ and U is a matrix whose columns are the eigenvectors associated with the k nonzero eigenvalues of $X'X$.

The condition indices of X matrix are defined as

$$\eta_j = \frac{\mu_{\max}}{\mu_j},\ j = 1, 2, ..., k$$

where $\mu_{\max} = Max(\mu_1, \mu_2, ..., \mu_k).$

If $\lambda_1, \lambda_2, ..., \lambda_k$ are the eigenvalues of $X'X$ then

$$X'X = (UDV')'UDV' = VD^2V' = V\Lambda V',$$

so $\mu_j^2 = \lambda_j,\ j = 1, 2, ..., k.$

Note that with $\mu_j^2 = \lambda_j$,

$$Var(b_j) = \sigma^2 \sum_{i=1}^{k} \frac{v_{ji}^2}{\mu_i^2}$$

$$VIF_j = \sum_{i=1}^{k} \frac{v_{ji}^2}{\mu_i^2}$$

$$p_{ij} = \frac{(v_{ij}^2 / \mu_i^2)}{VIF_j}.$$

The ill-conditioning in X is reflected in the sizes of the singular values. There will be a single small value for each non-linear dependency. The extent of ill-conditioning is described by how small is μ_j relative to $\mu_{\max}$.

It is suggested that the explanatory variables should be scaled to unit length, but should not be centered when computing p_{ij}. This will help in diagnosing the role of the intercept term in near-linear dependence. No unique guidance is available in the literature on the issue of centering the explanatory variables. The centering makes the intercept orthogonal to the explanatory variables. So this may remove the ill-conditioning due to the intercept term in the model.

Remedies for multicollinearity

Various techniques have been proposed to address problems arising from multicollinearity in the data.

1. Obtain more data

The harmful multicollinearity arises essentially because the rank of $X'X$ falls below k and $|X'X|$ is close to zero. Additional data may help in reducing the sampling variance of the estimates. The data needs to be collected so as to break the multicollinearity.

It is not always possible to collect additional data for various reasons, as follows.

- The experiment and process have finished and are no longer available.
- The economic constraints may also not allow for collecting the additional data.
- The additional data may not match the earlier collected data and may be unusual.
- If the data is in a time series, then a longer time series may force the ignoring of data that is too far in the past.

- If multicollinearity is due to any identity or exact relationship, then increasing the sample size will not help.
- Sometimes, it is not advisable to use the data even if it is available. For example, if data on consumption patterns are available for the years 1950-2010, one may not want to use them, as consumption patterns usually do not remain the same over such a long period.

2. Drop some variables that are collinear

If possible, identify the variables which seem to cause multicollinearity. These collinear variables can be dropped so as to match the condition of full rank of $X-$ matrix. The process of omitting the variables will be carried out on the basis of some kind of ordering of explanatory variables, e.g., those variables can be deleted first which have a smaller value of t-ratio. In another example, suppose the experimenter is not interested in all the parameters. In such cases, one can obtain estimators of the parameters of interest with smaller mean squared errors than the variance of OLSE by dropping some variables.

Eliminating some variables may reduce the model's predictive power. Sometimes there is no assurance that how the model will exhibit lower multicollinearity.

3. Use some relevant prior information

One may search for some relevant prior information about the regression coefficients. This may lead to the specification of estimates of some coefficients. The more general situation includes the specification of some exact linear restrictions and stochastic linear restrictions. Procedures such as restricted and mixed regression can be used for this purpose. The relevance and accuracy of information play an important role in such analysis, but ensuring them in practice is challenging. For example, the estimates derived in the U.K. may not be valid in India.

4. Employ generalized inverse

If $\text{rank}(X'X)<k$, then the generalized inverse can be used to find the inverse of $X'X$. Then β can be estimated by $\hat{\beta}=(X'X)^{-}X'y.$

In such a case, the estimates will not be unique except in the case of the use of the Moore-Penrose inverse of $(X'X)$. Different methods of finding generalized inverse may give different results. So applied workers will get different results. Moreover, it is also not known which method of finding generalized inverse is optimum.

5. Use of principal component regression

Principal component regression is based on principal component analysis. The k explanatory variables are transformed into a new set of orthogonal variables called principal components. Usually, this technique is used to reduce data dimensionality by retaining some variability in the explanatory variables, as reflected in the variability of the study variable. The principal components involve determining a set of linear combinations of the explanatory variables that capture the system's total variability and are mutually independent. The principal components obtained are ranked in the order of their importance. Importance is judged by the proportion of variability explained by a principal component relative to the system's total variability. The procedure then involves eliminating some of the principal components that explain relatively little variation. After eliminating the least important principal components, multiple regression is used by replacing the explanatory variables with the principal components. Then, the study variable is regressed on the selected principal components using ordinary least squares. Since all the principal components are orthogonal and therefore mutually independent, so OLS can be used without any problems. Once the estimates of regression coefficients for the reduced set of orthogonal variables (principal components) have been obtained, they are mathematically transformed into a new set of estimated regression coefficients corresponding to the original correlated set of variables. These new estimated coefficients are principal component estimators of the regression coefficients.

Suppose there are k explanatory variables $X_1, X_2, ..., X_k$. Consider the linear function of $X_1, X_2, .., X_k$ like

$$Z_1 = \sum_{i=1}^{k} a_i X_i$$

$$Z_2 = \sum_{i=1}^{k} b_i X_i \text{ etc.}$$

The constants $a_1, a_2, ..., a_k$ are determined such that the variance of Z_1 is maximized subject to the normalizing condition that $\sum_{i=1} a_i^2 = 1$. The constant $b_1, b_2, ..., b_k$ are determined such that the variance of Z_2 is maximized subject to the normality condition that $\sum_{i=1} b_i^2 = 1$ and is independent of the first principal component.

We continue with this process and obtain k such linear combinations, so that they are orthogonal to their preceding linear combinations and satisfy the normality condition. Then we obtain their variances. Suppose such linear combinations are $Z_1, Z_2, .., Z_k$ and for them, $Var(Z_1) > Var(Z_2) > ... > Var(Z_k)$. The linear

combination having the largest variance is the first principal component. The linear combination having the second-largest variance is the second-largest principal component and so on. These principal components have the property that $\sum_{i=1}^{k} Var(Z_i) = \sum_{i=1}^{k} Var(X_i)$. Also, the $X_1, X_2, ..., X_k$ are correlated but $Z_1, Z_2, .., Z_k$ are orthogonal or uncorrelated. So there will be zero multicollinearity among $Z_1, Z_2, .., Z_k$.

The problem of multicollinearity arises because $X_1, X_2, ..., X_k$ are not independent. Since the principal components based on $X_1, X_2, ..., X_k$ are mutually independent, so they can be used as explanatory variables, and such a regression will combat the multicollinearity.

Let $\lambda_1, \lambda_2, ..., \lambda_k$ be the eigenvalues of $X'X$, $\Lambda = diag(\lambda_1, \lambda_2, ..., \lambda_k)$ is $k \times k$ diagonal matrix, V is a $k \times k$ orthogonal matrix whose columns are the eigenvectors associated with $\lambda_1, \lambda_2, ..., \lambda_k$. Consider the canonical form of the linear model

$$\begin{aligned} y &= X\beta + \varepsilon \\ &= XVV'\beta + \varepsilon \\ &= Z\alpha + \varepsilon \end{aligned}$$

where $Z = XV$, $\alpha = V'\beta$, $V'X'XV = Z'Z = \Lambda$.

Columns of $Z = (Z_1, Z_2, ..., Z_k)$ define a new set of explanatory variables, which are called **principal components.**

The OLSE of α is

$$\begin{aligned} \hat{\alpha} &= (Z'Z)^{-1}Z'y \\ &= \Lambda^{-1}Z'y \end{aligned}$$

and its covariance matrix is

$$\begin{aligned} V(\hat{\alpha}) &= \sigma^2 (Z'Z)^{-1} \\ &= \sigma^2 \Lambda^{-1} \\ &= \sigma^2 diag\left(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, ..., \frac{1}{\lambda_k}\right) \end{aligned}$$

Note that λ_j is the variance of jth principal component and $Z'Z = \sum_{i=1}^{k}\sum_{j=1}^{k} Z_i Z_j = \Lambda$. A small eigenvalue of $X'X$ means that the linear relationship between the original explanatory variable exists, and the variance of the corresponding orthogonal regression coefficient is large, indicating that multicollinearity exists. If one or more λ_j is small, then it indicates that multicollinearity is present.

Retainment of principal components

The new set of variables, i.e., principal components, is orthogonal and retains the same variance as the original set. If multicollinearity is severe, then at least one eigenvalue will be small. Eliminating one or more principal components corresponding to the smallest eigenvalues will reduce the total variance in the model. Moreover, the principal components responsible for multicollinearity will be removed, thereby appreciably improving the resulting model.

The principal component matrix $Z = [Z_1, Z_2, ..., Z_k]$ with $Z_1, Z_2, ..., Z_k$ contains precisely the same information as the original data in X in the sense that the total variability in X and Z is the same. The difference between them is that the original data are arranged into a set of new variables that are uncorrelated with one another and can be ranked by the magnitudes of their eigenvalues. The jth column vector Z_j corresponding to the largest λ_j accounts for the largest proportion of the variation in the original data. Thus the Z_j's are indexed so that $\lambda_1 > \lambda_2 > ... > \lambda_k > 0$ and λ_j is the variance of Z_j.

The strategy of principal component elimination begins by discarding the component associated with the smallest eigenvalue. The idea behind doing so is that the principal component with the smallest eigenvalue contributes the least variance and is therefore the least informative.

Using this procedure, principal components are eliminated until the remaining components explain some preselected variance in terms of the percentage of the total variance. For example, if 90% of the total variance is needed, and suppose r principal components are eliminated, which means that $(k-r)$ principal components contribute 90% variation, then r is selected to satisfy

$$\frac{\sum_{i=1}^{k-r} \lambda_i}{\sum_{i=1}^{k} \lambda_i} > 0.90.$$

Various strategies to choose the required number of principal components are also available in the literature.

Suppose after using such a rule, the r principal components are eliminated. Now only $(k-r)$ components will be used for regression. So Z matrix is partitioned as

$$Z = (Z_r \quad Z_{k-r}) = X(V_r \quad V_{k-r})$$

where Z_r submatrix is of order $n \times r$ and contains the principal components to be eliminated. The submatrix Z_{k-r} is of order $n \times (k-r)$ and contains the principal components to be retained.

The reduced model obtained after the elimination of r principal components can be expressed as

$$y = Z_{k-r}\alpha_{k-r} + \varepsilon^*.$$

The random error component is represented as ε^* just to distinguish from ε. The reduced coefficients contain the coefficients associated with retained $Z_j's$. So

$$Z_{k-r} = (Z_1, Z_2, ..., Z_{k-r})$$
$$\alpha_{k-r} = (\alpha_1, \alpha_2, ..., \alpha_{k-r})$$
$$V_{k-r} = (V_1, V_2, ..., V_{k-r}).$$

Using OLS on the model with retained principal components, the OLSE of α_{k-r} is

$$\hat{\alpha}_{k-r} = (Z_{k-r}'Z_{k-r})^{-1}Z_{k-r}'y .$$

Now it is transformed back to the original explanatory variables as follows:

$$\alpha = V'\beta$$
$$\alpha_{k-r} = V_{k-r}'\beta$$
$$\Rightarrow \hat{\beta}_{pc} = V_{k-r}\hat{\alpha}_{k-r}$$

is the **principal component regression estimator** of β.

This method improves efficiency and reduces multicollinearity.

6. Ridge regression

The OLSE is the best linear unbiased estimator of the regression coefficient in the sense that it has minimum variance in the class of linear and unbiased estimators. However, if the condition of unbiasedness can be relaxed, then it is possible to find a biased estimator of the regression coefficient, say $\hat{\beta}$ that has a smaller variance than the unbiased OLSE b. The mean squared error (*MSE*) of $\hat{\beta}$ is

$$\begin{aligned} MSE(\hat{\beta}) &= E(\hat{\beta} - \beta)^2 \\ &= E\left[\left\{\hat{\beta} - E(\hat{\beta})\right\} + \left\{E(\hat{\beta}) - \beta\right\}\right]^2 \\ &= Var(\hat{\beta}) + \left[E(\hat{\beta}) - \beta\right]^2 \\ &= Var(\hat{\beta}) + \left[Bias(\hat{\beta})\right]^2. \end{aligned}$$

Thus $MSE(\hat{\beta})$ can be made smaller than $Var(\hat{\beta})$ by introducing a small bias is $\hat{\beta}$. One approach to do so is ridge regression. The ridge regression estimator is obtained by solving the normal equations of least squares estimation. The normal equations are modified as

$$(X'X+\delta I)\hat{\beta}_{ridge} = X'y$$
$$\Rightarrow \hat{\beta}_{ridge} = (X'X+\delta I)^{-1}X'y$$

is the **ridge regression estimator** of β and $\delta \geq 0$ is any characterizing scalar termed as **biasing parameter**. As $\delta \to 0, \hat{\beta} \to b(OLSE)$ and as $\delta \to \infty$, $\hat{\beta} \to 0$.

The larger the value of δ, larger shrinkage towards zero. Note that the OLSE is inappropriate because it has very high variance when multicollinearity is present in the data. On the other hand, a very small value of $\hat{\beta}$ may tend to accept the null hypothesis $H_0 : \beta = 0$ indicating that the corresponding variables are not relevant. The value of the biasing parameter controls the amount of shrinkage in the estimates.

Bias of the ridge regression estimator

The bias of $\hat{\beta}_{ridge}$ is

$$\begin{aligned} Bias(\hat{\beta}_{ridge}) &= E(\hat{\beta}_{ridge}) - \beta \\ &= (X'X+\delta I)^{-1}X'E(y) - \beta \\ &= \left[(X'X+\delta I)^{-1}X'X - I\right]\beta \\ &= (X'X+\delta I)^{-1}\left[X'X - X'X - \delta I\right]\beta \\ &= -\delta(X'X+\delta I)^{-1}\beta. \end{aligned}$$

Thus, the ridge regression estimator is a biased estimator of β.

Covariance matrix

The covariance matrix of $\hat{\beta}_{ridge}$ is defined as

$$V(\hat{\beta}_{ridge}) = E\left[\left\{\hat{\beta}_{ridge} - E(\hat{\beta}_{ridge})\right\}\left\{\hat{\beta}_{ridge} - E(\hat{\beta}_{ridge})\right\}'\right].$$

Since

$$\begin{aligned} \hat{\beta}_{ridge} - E(\hat{\beta}_{ridge}) &= (X'X+\delta I)^{-1}X'y - (X'X+\delta I)^{-1}X'X\beta \\ &= (X'X+\delta I)^{-1}X'(y - X\beta) \\ &= (X'X+\delta I)^{-1}X'\varepsilon, \end{aligned}$$

so

$$\begin{aligned} V(\hat{\beta}_{ridge}) &= (X'X+\delta I)^{-1}X'V(\varepsilon)X(X'X+\delta I)^{-1} \\ &= \sigma^2(X'X+\delta I)^{-1}X'X(X'X+\delta I)^{-1}. \end{aligned}$$

Mean squared error

The mean squared error of $\hat{\beta}_{ridge}$ is

$$\begin{aligned} MSE(\hat{\beta}_{ridge}) &= Var(\hat{\beta}_{ridge}) + \left[bias(\hat{\beta}_{ridge}) \right]^2 \\ &= tr\left[V(\hat{\beta}_{ridge}) \right] + \left[bias(\hat{\beta}_{ridge}) \right]^2 \\ &= \sigma^2 tr\left[(X'X + \delta I)^{-1} X'X (X'X + \delta I)^{-1} \right] + \delta^2 \beta' (X'X + \delta I)^{-2} \beta \\ &= \sigma^2 \sum_{j=1}^{k} \frac{\lambda_j}{(\lambda_j + \delta)^2} + \delta^2 \beta' (X'X + \delta I)^{-2} \beta \end{aligned}$$

where $\lambda_1, \lambda_2, ..., \lambda_k$ are the eigenvalues of $X'X$.

Thus as δ increases, the bias in $\hat{\beta}_{ridge}$ increases, but its variance decreases. Thus, the trade-off between bias and variance hinges upon the value of δ. It can be shown that there exists a value of δ such that

$$MSE(\hat{\beta}_{ridge}) < Var(b)$$

provided $\beta'\beta$ is bounded.

Choice of δ

The estimation of the ridge regression estimator depends upon the value of δ. Various approaches have been suggested in the literature to determine the value of δ. The value of δ can be chosen on the bias of criteria like

- the stability of estimators with respect to δ.
- reasonable signs.
- the magnitude of residual sum of squares, etc.

We consider here the determination of δ by the inspection of the ridge trace.

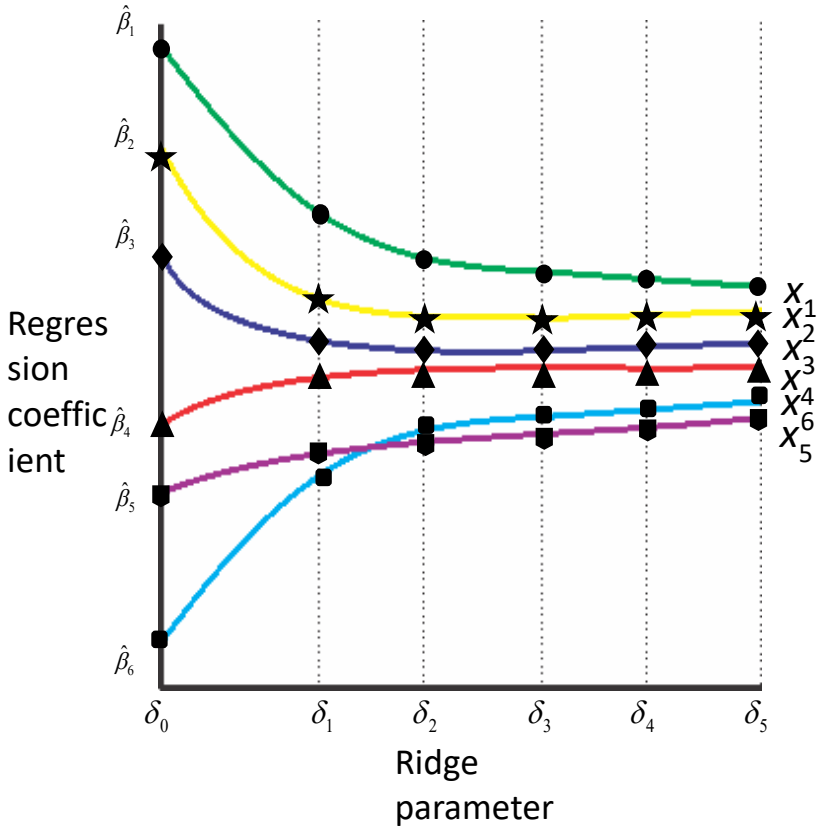
Ridge trace

Ridge trace is the graphical display of the ridge regression estimator versus δ.

If multicollinearity is severe, instability in regression coefficients is reflected in the ridge trace. As δ increases, some of the ridge estimates vary dramatically, and they stabilize at some value of δ. The objective in ridge trace is to inspect the trace (curve) and find the reasonable small value of δ at which the ridge regression estimators are stable. The ridge regression estimator with such a choice of δ will have a smaller MSE than the variance of OLSE.

An example of a ridge trace for a model with 6 parameters is as follows:

In this ridge trace, the $\hat{\beta}_{ridge}$ is evaluated for various choices of δ and the corresponding values of all regression coefficients $\hat{\beta}_{j(ridge)}$'s, j=1,2,…,6 are plotted versus δ. These values are denoted by different symbols and are joined by a smooth curve. This produces a ridge trace for the respective parameter. Now choose the value of δ where all the curves stabilize and become nearly parallel. For example, the curves in the following figure become almost parallel, starting from $\delta = \delta_4$ or so. Thus, one possible choice of δ is $\delta = \delta_4$ and parameters can be estimated as $\hat{\beta}_{ridge} = \left(X'X + \delta_4 I\right)^{-1} X'y$.



The figure clearly reveals multicollinearity in the data. The behaviour of $\hat{\beta}_{i(ridge)}$ at $\delta_0 \approx 0$ is very different from other values of δ. For small values of δ, the estimates change rapidly. The estimates stabilize gradually as δ increases. The value of δ at which all the estimates stabilize gives the desired value of δ because moving away from such δ will not bring any appreciable reduction in the residual sum of squares. It multicollinearity is present, then the variation in ridge regression estimators is rapid around $\delta = 0$. The optimal δ is chosen such that after that value of δ, almost all traces stabilize.

Limitations

1. The choice of δ is data-dependent and therefore is a random variable. Using it as a random variable violates the assumption that δ is constant. This will disturb the optimal properties derived under the assumption of the constancy of δ.
2. The value of δ lies in the interval $(0,\infty)$. So a large number of values are required for exploration. This result is a waste of time. This is not a big issue when working with the software.
3. The choice of δ from graphical display may not be unique. Different people may choose different δ and consequently, the values of ridge regression estimators will change. Another choice of δ is $\delta = \frac{k\hat{\sigma}^2}{b'b}$ where b and $\hat{\sigma}^2$ are obtained from the least-squares estimation.
4. The stability of numerical estimates of $\hat{\beta}_i's$ is a rough way to determine δ. Different estimates may exhibit stability for different δ, and it may often be hard to strike a compromise. In such a situation, generalized ridge regression estimators are used.
5. There is no guidance available regarding the testing of hypotheses and for confidence interval estimation.

Idea behind the ridge regression estimator

The problem of multicollinearity arises because some of the eigenvalues of $X'X$ are close to zero or are zero. So if $\lambda_1, \lambda_2, ..., \lambda_p$ are the characteristic roots, and if

$$X'X = \Lambda = diag(\lambda_1, \lambda_2, ..., \lambda_k)$$

then

$$\hat{\beta}_{ridge} = (I + \delta\Lambda^{-1})^{-1} b$$

where b is the OLSE of β given by

$$b = (X'X)^{-1}X'y = \Lambda^{-1}X'y.$$

Thus, a particular element will be of the forms

$$\frac{1}{1+\frac{\delta}{\lambda_i}} b_i = \frac{\lambda_i}{\lambda_i + \delta} b_i.$$

So a small quantity δ is added to λ_i so that if $\lambda_i = 0,$ even then $\frac{\lambda_i}{\lambda_i + \delta}$ remains meaningful.

Another interpretation of the ridge regression estimator

In the model $y = X\beta + \varepsilon,$ obtain the least squares estimator of β when $\sum_{i=1}^{k} \beta_1^2 = C,$ where C is some constant. So minimize

$$\delta(\beta) = (y - X\beta)'(y - X\beta) + \delta(\beta'\beta - C)$$

where δ is the Lagrangian multiplier. Differentiating $S(\beta)$ with respect to β, the normal equations are obtained as

$$\frac{\partial S(\beta)}{\partial \beta} = 0 \Rightarrow -2X'y + 2X'X\beta + 2\delta\beta = 0$$

$$\Rightarrow \hat{\beta}_{ridge} = (X'X + \delta I)^{-1} X'y.$$

Note that if C is very small, it may indicate that most regression coefficients are close to zero, and if C is large, it may indicate that the regression coefficients are far from zero. So C imposes a penalty on the regression coefficients to enable their estimation.